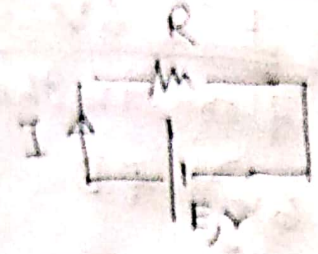
Solution of Current electricity
Problem set 1

1.



$$I = \frac{E}{R+r}$$

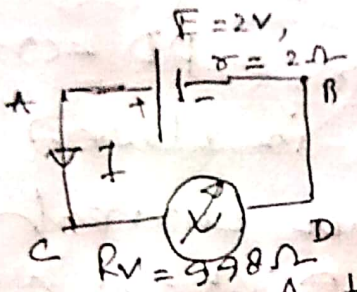
$$0.5 = \frac{10}{R+3} \Rightarrow \frac{1}{2} = \frac{10}{R+3}$$

$$\Rightarrow R = 17\,\Omega$$

∵ current flows in the diff. of e.m.f so PD across cell

$$V = E - Ir$$
$$= 10 - \frac{1}{2}\times 3 = 10 - 1.5$$
$$= 8.5\,V$$

2.



Voltmeter, here, is used to measure e.m.f of cell.

So e.m.f to be measured $\boxed{E = 2V}$

Since the cell is in closed circuit & sends current in the diff. of its emf

So PD across terminals of cell

$= V_A - V_B = \boxed{E - Ir} = V_C - V_D = IR$

reading of voltmeter. $\boxed{(IR_v)}$

Now $I = \frac{E}{R_v + r} = \frac{2}{998+2} = \frac{2}{1000}\,A$

∴ Reading of voltmeter $V_C - V_D = IR_v$ (OR) $E - Ir$

$$V_C - V_D = IR_v = \frac{2}{1000}\times 998$$
$$= 1.996\,V$$

error in measurement

$$= \frac{2 - 1.996}{2}\times 100 = \frac{.004}{2}\times 100$$
$$= .002\times 100$$
$$= 0.2\%$$

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(3) PD across a cell in open circuit ($I = 0$)

= e.m.f of cell = $E = 6V$

& when current I (2A) flows in the direction of e.m.f ($E = 6V$) PD across cell $= E - Ir = 4V$

$$6 - 2 \times r = 4$$

$$2r = 2 \quad \boxed{r = 1\Omega}$$

(4) $E = 1.8V$, $I = 0.5A$.

$E - Ir = 1.6$ $\quad$ $E - Ir$ = PD across a cell in closed circuit when current flows in the dir. of e.m.f.

$1.8 - \frac{1}{2} \times 0.5$

$$E - 0.5r = 1.8 \quad \text{—(1)}$$

$$E - 1r = 1.6 \quad \text{—(2)}$$

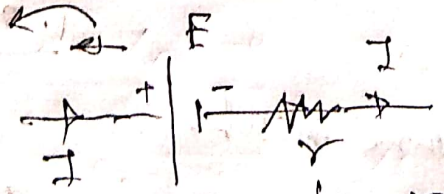
$$0.5r = 0.2$$

$$\frac{1}{2}r = 0.2 \quad \boxed{r = 0.4\Omega}$$

$$E - r = 1.6 \Rightarrow E - 0.4 = 1.6$$

$$\boxed{E = 2V}$$

(5) dir. of e.m.f



When a cell is recharged, 'I' flows in opposition to 'E' ie from +ve to −ve terminal inside the battery.

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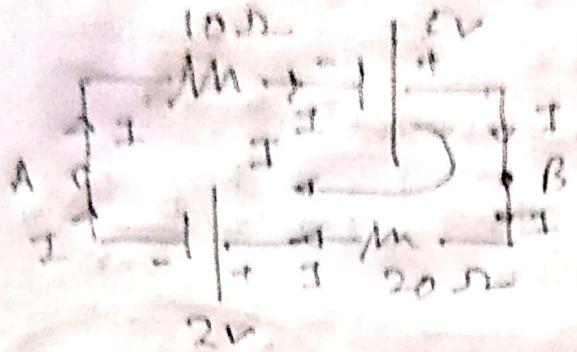
1D across terminals of all

$V = I + I + 1 = 2 + 1 = (0.1)$

$= 0.2$ volt

(d)



cells and resistors are in series.

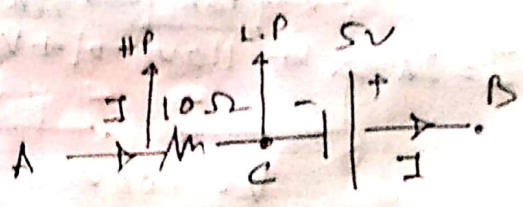
∴ cells oppose each other (their similar polarity terminals are connected together) So net emf = 5 − 2 = 3V

& eqn resistance = 10 + 20 = 30Ω.

$\therefore I = \frac{E_{net}}{R_{net}} = \frac{3}{30} = \frac{1}{10}$ A.

Now $V_A - V_B$?

Consider upper row



HP = high potential
LP = low potential

$V_A - V_B = (V_A - V_C) + (V_C - V_B)$

$= I \times 10 - (V_B - V_C)$

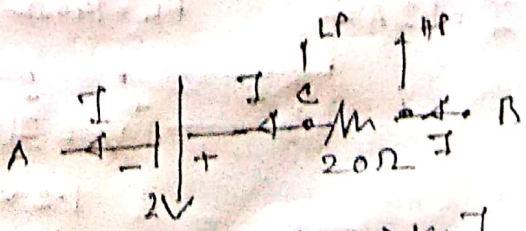
$= \frac{1}{10} \times 10 - 5$

$= 1 - 5 = -4V$

$\therefore V_A < V_B$ or $V_B - V_A = 4V$

OR

Consider lower row



$V_A - V_B = (V_A - V_C) + (V_C - V_B)$

$= -(V_C - V_A) - (V_B - V_C)$ [∵ $V_B > V_C$ & $V_C > V_A$]

$= -2 - I \times 20 = -2 - \frac{1}{10} \times 20 = -4V$

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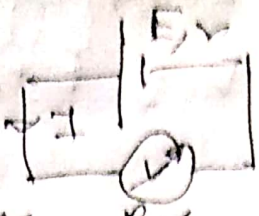
7. Same as Problem (2)



PD to be measured $= E$

PD measured by voltmeter $= E - Ir$

So error in measurement $= Ir$

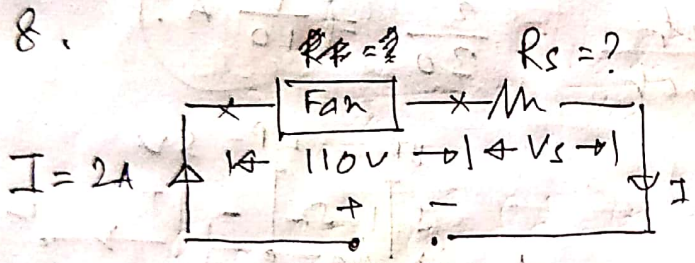
$$\% \text{ error} = \frac{Ir}{E} \times 100 = \left(\frac{E}{R_v + r}\right) \times \frac{r}{E} \times 100$$

$$= \frac{r}{R_v + r} \times 100$$

Now $\frac{r}{R_v + r} \times 100 = 20$

$$\frac{5}{R_v + 5} = \frac{1}{5} \Rightarrow R_v = 20\Omega$$

8.



PD acrss Fan + PD across 'Rs' = V

$110 + V_s = 200$

$V_s = 90$ volt

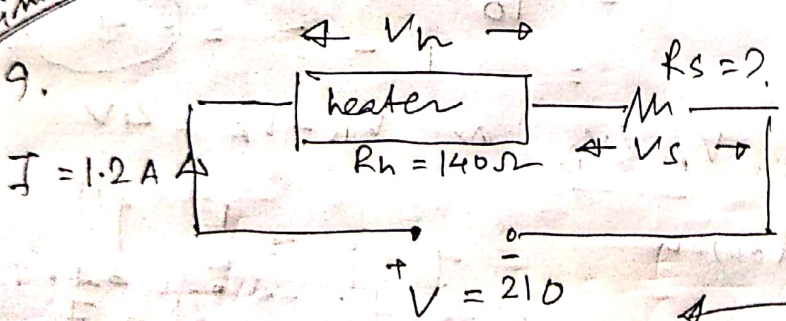
$I \times R_s = 90$

$2 \times R_s = 90$

$R_s = 45\Omega$

Similar

9.



$V_h + V_s = V$

$I \times R_h + I \times R_s = V$

$1.2 \times 140 + 1.2 \times R_s = 210$

$\Rightarrow R_s = 35\Omega$

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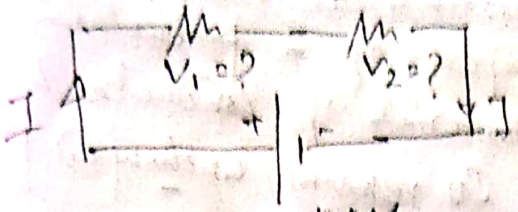
$R_1 = 12\Omega$ $R_2 = 4\Omega$



$V = 40V$

$r = 0$

$V = IR$

In series I = const

$\therefore V \propto R$

$\therefore V_1 : V_2 = 12 : 4$

$V_1 : V_2 = 3 : 1$

$[\& \ V = V_1 + V_2 = 40V]$

So $V_1 = \left(\frac{3}{3+1}\right) \times 40 = \underline{\underline{30V}}$

$V_2 = \left(\frac{1}{3+1}\right) \times 40 = \underline{\underline{10V}}$

OR

$I = \frac{V}{R_1 + R_2} = \frac{40}{12+4} = \frac{\cancel{40}\ 5}{\cancel{16}\ 2}$

$= \frac{5}{2} A$

$\therefore V_1 = IR_1$

$= 5/2 \times 12 = \underline{\underline{30V}}$

$V_2 = IR_2 = \frac{5}{2} \times 4 = \underline{\underline{10V}}$

(11) $R = \frac{\rho l}{A} = \frac{\rho l A}{A^2} = \frac{\rho V}{\left(\frac{\pi d^2}{4}\right)^2}$ $\therefore \boxed{R \propto \frac{1}{d^4}}$

$\therefore \frac{R_A}{R_B} = \left(\frac{d_B}{d_A}\right)^4 = \frac{2^4}{1^4} = \frac{16}{1}$

(12) $R = \frac{\rho l}{A}$

$R = \frac{\rho l}{\frac{\pi d^2}{4}}$

$R \propto \frac{l}{d^2}$

$\therefore \frac{R_2}{R_1} = \left(\frac{l_2}{l_1}\right) \times \left(\frac{d_1}{d_2}\right)^2$

$= \left(\frac{4l}{l}\right) \times \left(\frac{d}{2d}\right)^2$

$= 4 \times \frac{1}{4} = 1$

$\therefore R_2 = R_1 = R$

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(13) $R = \frac{\rho l}{A} = \frac{\rho l^2}{\rho A}$ [Resistivity is intrinsic property of a particular material of conductor so remains unchanged]

$R = \frac{\rho l^2}{\text{volume}}$

$\because \boxed{R \propto l^2}$ $\quad \frac{R_f}{R_i} = \left(\frac{l_f}{l_i}\right)^2 = \left[\left(\frac{110}{100}\right)\frac{l_i}{l_i}\right]^2$

$$\frac{R_f}{R_i} = \frac{121}{100}$$

$$\text{or, } \frac{R_f}{R_i} - 1 = \frac{121}{100} - 1 \Rightarrow \frac{R_f - R_i}{R_i} = \frac{21}{100}$$

$$\text{or } \left(\frac{R_f - R_i}{R_i}\right) \times 100 = \frac{21}{100} \times 100 = 21\%$$

(14) $R = \frac{\rho l}{A}$ $\quad [m = Al\rho, \; A = \frac{m}{l\rho}]$

$= \frac{\rho l}{\left(\frac{m}{l\rho}\right)} = \frac{\rho d l^2}{m}$ $\quad A$ = cross section, l = length, d = density

$\therefore \boxed{R \propto \frac{l^2}{m}}$ $\quad (\because \rho d = \text{const})$

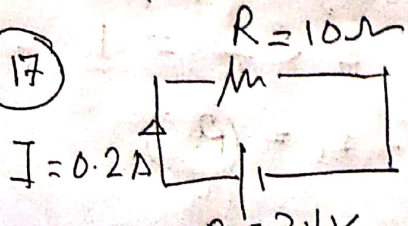
$\therefore R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3}$

$= \frac{(5)^2}{(1)} : \frac{(3)^2}{(3)} : \frac{(1)^2}{(5)}$

$= 25 : 3 : \frac{1}{5}$

$= \boxed{125 : 15 : 1}$

(17)



$I = \frac{E}{R+r} \Rightarrow 0.2 = \frac{2.1}{10+r}$

$\Rightarrow 2 + 0.2r = 2.1$

$0.2r = 0.1$ $\quad \boxed{r = 0.5\Omega}$

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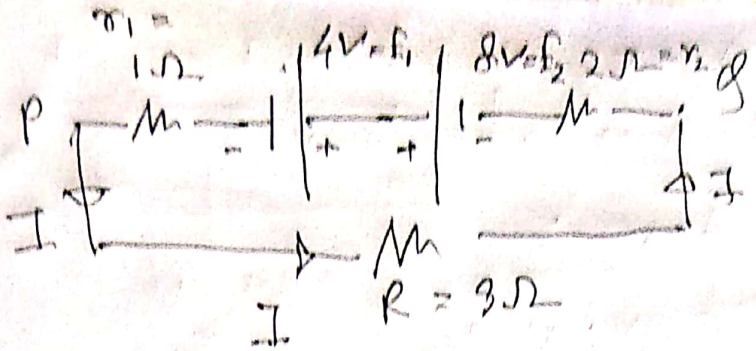
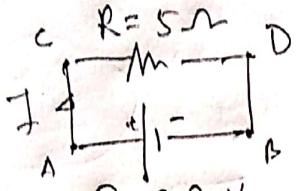


$$I = \frac{E_2 - E_1}{R + r_1 + r_2} = \frac{8-4}{9+1+2} = \frac{4}{12} = \frac{1}{3} A$$

$$V_P - V_Q = IR = \frac{1}{3} \times 9 = 3V$$

16 $E = 2.2V$ & $E - Ir = 1.8 V$



$E = 2.2 V$

$r = ?$

$V_A - V_B = E - Ir$

$V_C - V_D = IR$

$\because V_A - V_B = V_C - V_D$

$\therefore E - Ir = IR$

$\therefore IR = 1.8$

$I = \frac{1.8}{5} A$

$$2.2 - \left(\frac{1.8}{5}\right) r = 1.8$$

$$\frac{1.8}{5} r = 0.4$$

$$r = \frac{2}{1.8} = \frac{20}{18} = \frac{10}{9} \Omega.$$

18 $\frac{E}{R+r} = I \rightarrow$ C-1 $\frac{E}{2+r} = 2$ —(i)

C-2 $\frac{E}{9+r} = 0.5$ —(ii)

(i) ÷ (ii) $\frac{9+r}{2+r} = \frac{2}{0.5}$

$$\frac{9+r}{2+r} = 4 \Rightarrow 9 + r = 8 + 4r$$

$$3r = 1$$

$$\boxed{r = \frac{1}{3} \Omega}$$

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